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10 ACADEMY · WEEK 0 · INTERIM REPORT

African Climate Trend Analysis

COP32 Preparatory EDA — Interim Submission

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GitHub: github.com/rediet-shewarega/climate-challenge-week0

Track: Data Engineering / Machine Learning Engineering

Date: April 27, 2026

Submission Deadline: April 26, 2026 — 8:00 PM UTC (Interim)

Task 1 — Git & Environment Setup

This section documents the complete version-control and development environment setup completed as the foundation for the Week 0 climate analysis challenge. All work lives in a public GitHub repository under the account **rediet-shewarega**, following the Conventional Commits specification and the branching strategy prescribed by the challenge.

1.1 Repository Overview

5	3	1
FEATURE BRANCHES	CONVENTIONAL COMMITS	PR MERGED TO MAIN

Repository URL: **https://github.com/rediet-shewarega/climate-challenge-week0**

1.2 Commit History on `setup-task`

```
dcd9c32  init: add .gitignore and project scaffold
Added .gitignore excluding data/, CSVs, venvs, and Jupyter checkpoints. Added
requirements.txt with pinned core dependencies. Added README.md documenting
environment setup and folder structure.
```

```
e57a40d  ci: add GitHub Actions workflow for dependency validation
Added .github/workflows/ci.yml — triggers on push to main and all feature branches,
installs requirements, and validates core imports on ubuntu-latest with Python 3.11.
```

```
cd24e8a  chore: scaffold project directories with init files and READMEs
Added src/, tests/, scripts/, notebooks/, app/ with __init__.py stubs and README
files per the challenge folder structure specification.
```

1.3 Branch Strategy

Branch	Purpose	Status
`main`	Production-ready, merged code	✅ Live
`setup-task`	Task 1 — Git & environment setup	✅ Merged via PR #1
`eda-ethiopia`	Task 2 — Ethiopia EDA notebook	✅ Pushed
`compare-countries`	Task 3 — Cross-country comparison	✅ Pushed
`dashboard-dev`	Bonus — Streamlit dashboard	✅ Pushed

1.4 GitHub Actions CI

A CI workflow (`.github/workflows/ci.yml`) was configured to run automatically on every push to `main` and all feature branches, and on every pull request. The workflow:

- ✅ Sets up Python 3.11 on ubuntu-latest
- ✅ Runs `python --version` to confirm environment
- ✅ Runs `pip install -r requirements.txt`
- ✅ Validates all core imports (pandas, numpy, matplotlib, seaborn, scipy, streamlit, plotly)

```
✓ Set up Python 3.11
✓ Display Python version → Python 3.11.x
✓ Install dependencies → pip install -r requirements.txt
✓ Check imports → All core imports OK
```

1.5 Folder Structure

```
climate-challenge-week0/
├── .github/workflows/ci.yml      ← GitHub Actions CI
├── .gitignore                    ← excludes data/, *.csv, venv/
├── requirements.txt
├── README.md
```

```
├── data/                    ← gitignored; raw & cleaned CSVs here
├── notebooks/
│   ├── ethiopia_eda.ipynb
│   ├── kenya_eda.ipynb
│   ├── sudan_eda.ipynb
│   ├── tanzania_eda.ipynb
│   ├── nigeria_eda.ipynb
│   └── compare_countries.ipynb
├── app/
│   ├── main.py              ← Streamlit dashboard
│   └── utils.py
├── scripts/
├── src/
└── tests/
```

1.6 Environment Reproduction

git clone repo → python -m venv venv → pip install -r requirements.txt → jupyter notebook

Full instructions are documented in `README.md` at the root of the repository, covering prerequisites, virtual environment creation (venv/conda), dependency installation, data file placement, notebook execution, and Streamlit dashboard launch.

Task 2 — Data Profiling & Cleaning Approach

This section outlines the analytical approach for Task 2: Data Profiling, Cleaning, and Exploratory Data Analysis. Notebooks have been structured and are ready to execute once the NASA POWER CSV files are placed in the `data/` directory. The approach is consistent across all five countries, with country-specific observations expected upon execution.

2.1 Countries & Dataset

🇪🇹 Ethiopia 🇰🇪 Kenya 🇸🇩 Sudan 🇹🇿 Tanzania 🇳🇬 Nigeria

Source: NASA POWER (Prediction of Worldwide Energy Resources) database. Daily observations, January 2015 – March 2026. Variables: T2M, T2M_MAX, T2M_MIN, T2M_RANGE, PRECTOTCORR, RH2M, WS2M, WS2M_MAX, PS, QV2M.

2.2 Six-Stage Pipeline

Stage 1 — Data Loading & Date Parsing

- Load CSV via `pd.read_csv()`
- Add `Country` column identifier
- Convert YEAR + DOY → proper datetime using `pd.to_datetime(YEAR*1000 + DOY, format="%Y%j")`
- Extract `Month` column for seasonal analysis

Stage 2 — Summary Statistics & Missing Values

- Replace NASA sentinel `-999` with `np.nan` before any stats
- Run `df.duplicated().sum()` and drop duplicates
- Run `df.describe()` with written interpretation
- Report columns with >5% missing values

Stage 3 — Outlier Detection & Cleaning

- Compute Z-scores for 7 key variables

Stage 4 — Time Series Analysis

- Monthly average T2M line chart (2015–2026) with warmest/coolest annotations

- Flag rows where |Z| > 3
- Decision: *retain* outliers (real extreme events for COP32 narrative)
- Forward-fill remaining NaNs; drop rows >30% missing
- Export to `data/<country>_clean.csv`

- OLS warming trend line (°C/year)
- Monthly total PRECTOTCORR bar chart with peak rainy season annotation
- Markdown interpretation of trends and anomalies

Stage 5 — Correlation & Relationships

- Lower-triangle correlation heatmap (all numeric columns)
- Scatter: T2M vs RH2M with regression line (r value)
- Scatter: T2M_RANGE vs WS2M
- Identify and interpret top 3 correlations

Stage 6 — Distribution Analysis

- PRECTOTCORR histogram on log(1+x) scale with skewness annotation
- Bubble chart: T2M vs RH2M, bubble size ∝ PRECTOTCORR
- Comment on distribution shape and climate interpretation

2.3 Key Analytical Decisions

ON OUTLIER TREATMENT

Extreme precipitation events that register |Z| > 3 are **retained** rather than dropped or capped. These represent climatically real extreme events — flash floods, El Niño-driven rainfall surges — which are precisely the signals relevant to a COP32 vulnerability analysis. Removing them would systematically understate climate risk.

ON MISSING VALUE STRATEGY

Forward-fill (`ffill()`) is applied to weather variables following NASA POWER convention, where short gaps typically represent sensor outages with negligible climatological change. Rows where more than 30% of numeric values remain missing after forward-fill are dropped to prevent distortion of statistical summaries.

ON THE -999 SENTINEL

NASA POWER uses `−999` as a sentinel value for missing or out-of-range measurements. This replacement is performed as the *first* operation after loading — before `describe()`, before Z-score computation, before any visualization — to ensure no statistics are contaminated by this artificial value.

2.4 Expected Insights Preview

Preliminary Hypotheses (to be validated with data)

- Ethiopia: bimodal Belg (Mar–May) and Kiremt (Jun–Sep) precipitation structure; 2020–2023 drought anomaly visible in Kiremt suppression
- Kenya: strong ENSO-driven precipitation variability; Indian Ocean Dipole influence on long rains (Mar–May)
- Sudan: highest mean temperatures; longest consecutive dry-day sequences; sharpest warming trend
- Tanzania: pronounced single rainy season (Nov–Apr); coastal vs. highland temperature gradient
- Nigeria: highest precipitation CV due to Sahel–forest latitudinal gradient; Harmattan dry-season signal in WS2M

2.5 Planned Deliverables for Final Submission

✅ 5 cleaned CSVs exported to `data/` (gitignored, not committed)

✅ 5 country EDA notebooks with all 6 stages complete and markdown interpretations

✅ `compare_countries.ipynb` with cross-country comparison and vulnerability ranking

✅ Streamlit dashboard deployed to Streamlit Community Cloud

✅ Final report in Medium blog style (PDF format)

2.6 References Consulted

WMO State of the Climate in Africa 2024 — used for regional warming baselines and subregional decomposition framework.
World Bank Climate Risk Country Profiles (Ethiopia, Kenya) — used as structural template for country-level analysis.
NASA POWER Data Access Viewer — primary data source documentation.

Addis Ababa Declaration (ACS2, Sep 2025) — rhetorical frame: Africa-as-strategic-player at COP32.

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